Housing Recovery and Reconstruction Platform-Nepal (HRRP)

Orientation Program for Partner organizations on

Correction/Exception Manual & New approved Housing Designs

Organized by:

National Reconstruction Authority(NRA)

MOUD/District Project Implementation Unit (DLPIU)

Housing Recovery and Reconstruction Platform-Nepal (HRRP)

Orientation Program

Time	Program
15 Minutes	Arrival/Registration/Opening Session
120 Minutes (2 hour)	Correction and Exceptions Manual for Masonry Structures and Updated Inspection Form • Objective of manual • Use of manual • Updated forms • Overview of training on manual for technical staff
15 Minutes	Tea Break
30 Minutes	Brief Introduction on New approved Housing Designs by MOUD/CLPIU
30 Minutes	Communication Skills & FAQs
45 Minutes	Discussion

Prevailing Situation

Wall Span more than 4.5 m(Non Compliant)

Housing Recovery & Reconstruction Platform
(HRRP)

Shape of Building (Non-Compliant)

Housing Recovery & Reconstruction Platform (HRRP)

Mixed Materials(Non Compliant)

Problem ?

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Is it Compliant ?

Housing Recovery & Reconstruction Platform (HRRP)

One Storey & Attic(Non Compliant)

What’s wrong in this ?

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Hybrid structure (Non Compliant)

Correction/ Exception Manual For Masonry Structure

Need

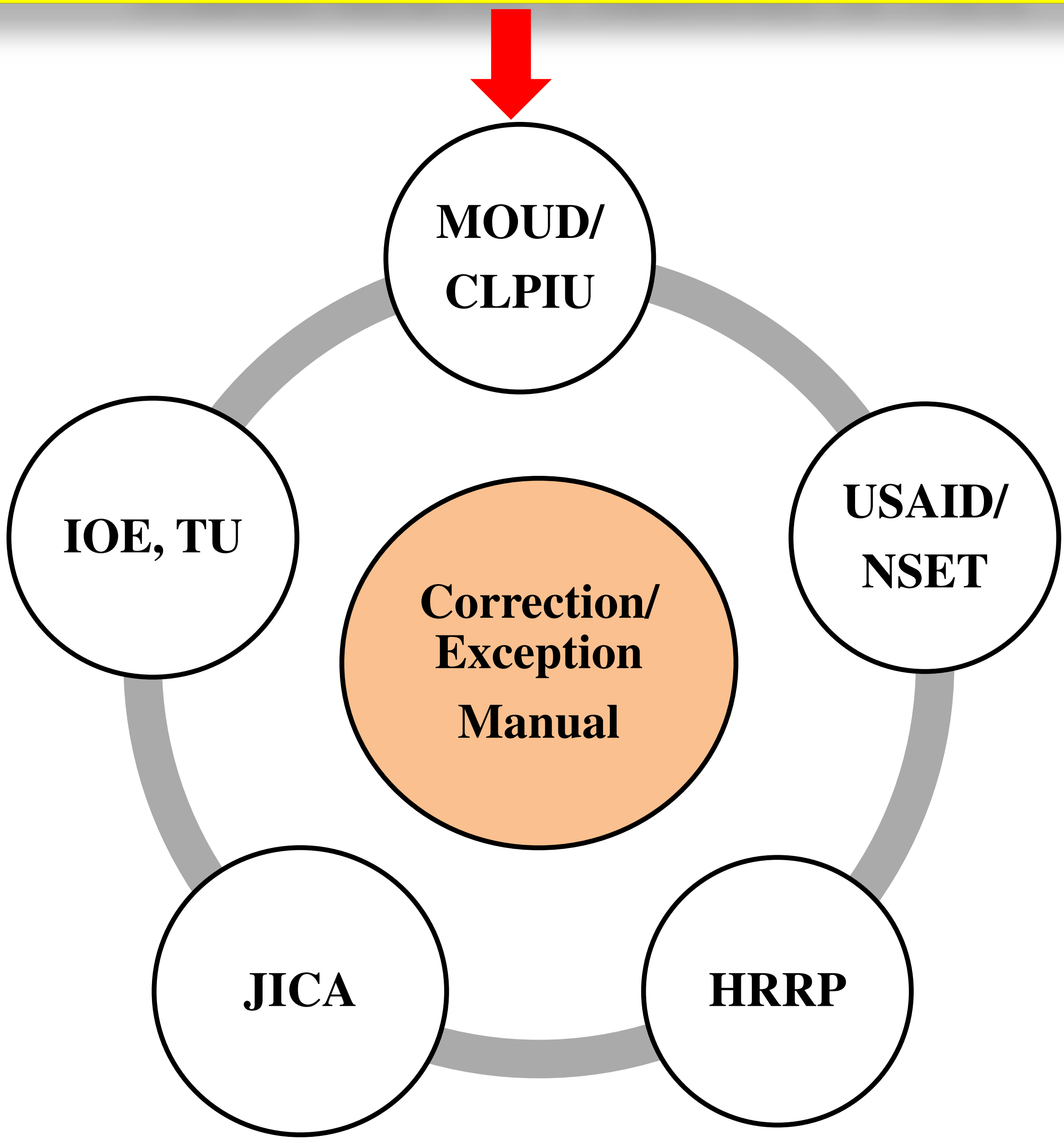
- Almost 7,17,251 household eligible to receive 300,000 NPRs reconstruction grant (52,211 Dolakha)
- Grant disbursed based on construction Compliance with the Minimum Requirements(MRs) as per the NBC & the inspection Checklists
- Almost 57,000 household started reconstruct houses
- Many houses have been found during Inspection to be non-compliant with MRs
- In order to bring these houses to compliance, corrective measures are required
- Non-compliant buildings are vulnerable to future Earthquakes

Objectives of Manual

- To introduce the exceptional cases, other than mentioned in Minimum Requirements(MRs)
- To provide correction measures along with their step by step Procedures
- To support households to overcome non-compliant issues and secure approval to receive tranches
- To support Engineers working for housing reconstruction program

Initiatives

Standardization Committee of NRA



Scope of Manual

Classification of Building, Building Act 2055

- **Category "A"**

Modern building to be built, based on the international state-of-the-art, also in pursuance of the building codes to be followed in developed countries.

- **Category "B"**

Buildings with plinth area of more than one Thousand square feet, with more than Three floors including the ground floor or with structural span of more than 4.5 meters.

- **Category "C"**

Buildings with plinth area of up to One Thousand square feet, with up to Three floors including the ground floor or with structural span of up to 4.5 meters.

- **Category "D"**

Small Houses, sheds made of baked or unbaked brick, Stone ,Clay, bamboos, grass etc., except those set forth clause (a), (b) & (c)

Applicability

- This Manual is applicable within certain limitations as guided by NBC 202 & NBC 203, which are based on NBC 105 Seismic design Code
- The correction and exceptions set forth in this manual are applicable only for the residential houses that have been constructed after Gorkha Earthquake 2015 Under GoN reconstruction program
- It intends to achieve the minimum acceptable structural safety envisioned in NBC 105
- The designs mentioned in the manual are ready-to-use designs for all structural component , but some provision are set as advisory measures

Limitations

- **Only covers load bearing masonry building**
- **The measures are for newly constructed, or under construction buildings**
- **Applicable for residential houses that have been constructed after Gorkha Earthquake 2015**
- **Only relevant for residential building fall under category 'C' and 'D' of NBC**

Contents of Manual

Part A:Exception/Correction

Part B: Mitigation Measures

Annex

Exception

- Exception is the cases of the buildings that do not comply with MRs but are structurally safe as per NBC 105 including the cases mentioned in NBC
- The exceptional cases were drafted by the NRA technical standardization committee on the basis of seismic requirements following NBC 105
- When all required exception measures have been completed the building can be approved for the subsequent tranche of the housing reconstruction grant

Correction

- **Correction is the corrective measures required to make newly constructed or under construction buildings compliant with the seismic resistance standard as per NBC 105**
- **The appropriate corrective measures can be carried out on any individual building which is missing earthquake resistant elements as per MRs and at any stage of construction. These measures were drafted by the NRA technical standardization committee on the basis of seismic requirements following NBC 105**
- **When all required corrective measures have been completed the building can be approved for the subsequent tranche of the housing reconstruction grant**

Part B: Mitigation Measures

- The mitigation measures consists of **step by step procedure** for implementing the correction.

- Depending upon the availability of materials and workmanship suitable mitigation measures can be adopted.

Components

Part A:Exception/Correction

Components	Details
1. Site Selection	• Site condition(treatment/ retaining wall
2. Shape of building	• Span of wall, size of room, Height of wall • Shape of building(Proportion) • Number of Storey
3. Materials	• Using improper materials, Mix materials
4. Foundation	• Insufficient Foundation
5. Vertical Member	• Vertical Member • Wooden Vertical Member
6. Plinth Beam	• No Plinth beam/ Level of Plinth

Part A:Exception/Correction

Components	Details
7. Walls	• Weak masonry/ Lack of through stones • Vulnerable gable wall
8. Door & Windows	• Inappropriate position and Size of Opening
9. Horizontal Band	• R.C. Horizontal Band • Wooden Horizontal Band
10. Roof	• Connection/ Heavy Materials for Roofing

1. Site Selection

Problems

- **Site selection shall be done so as to minimize risks in relation to natural hazards. No buildings shall be constructed in hazardous areas:**

Hazard Area	Exception/Correction
Geological fault or Ruptured area	• Not to be evaluated for residential building
Water –Logged area River bank	• Constructions is allowable, if the site is appropriately treated be undertaken . • Maintain minimum distance from river bank and observed high flood level.
Steep Slope	• If the terrain is stable and soil is medium to hard, construction on steep slopes is allowable.
Filled Area	• If a building is to be constructed on filled-ground, the foundation shall be deep enough so as to rest on the firm ground surface beneath the fill.
Rock Fall Area	• Building can be constructed in such areas except in risky rock fall area identified by geological study and local knowledge.

1. Site Selection Contd.

Solutions

- **Appropriate treatment of the Site**
- **Construction of retaining wall**

- **Steep slope area**

✓ **Constructed retaining wall**

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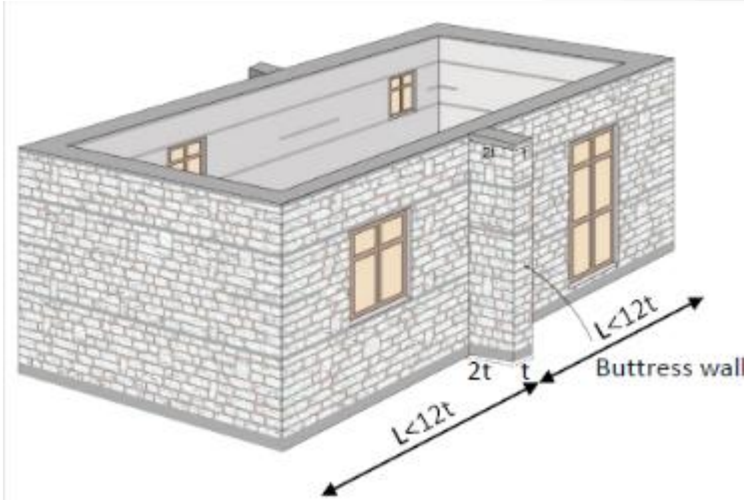
2.1 Span of wall, size of room, Height of Wall

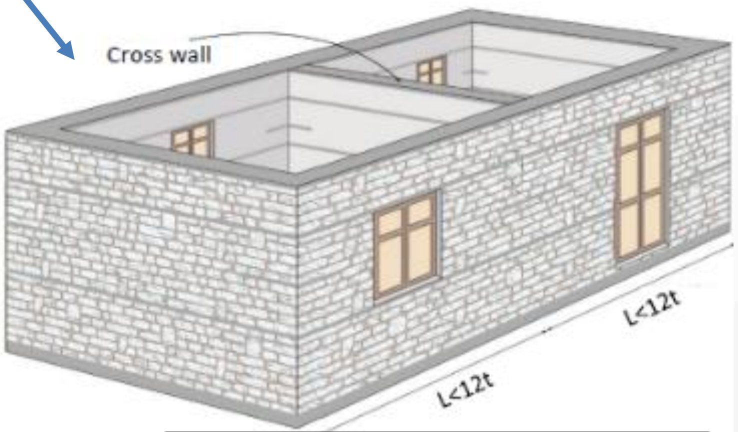
Span & size room is more than MRs

Option: 1



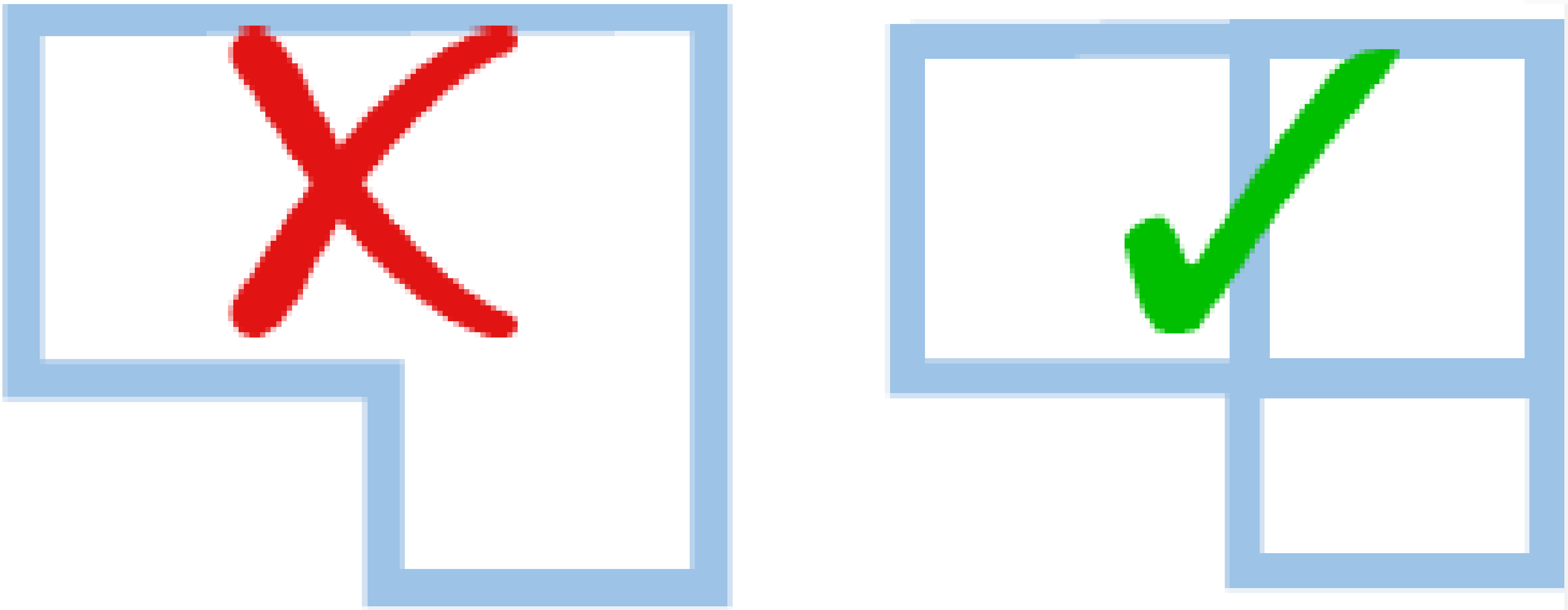
Adding buttress wall

Option: 2



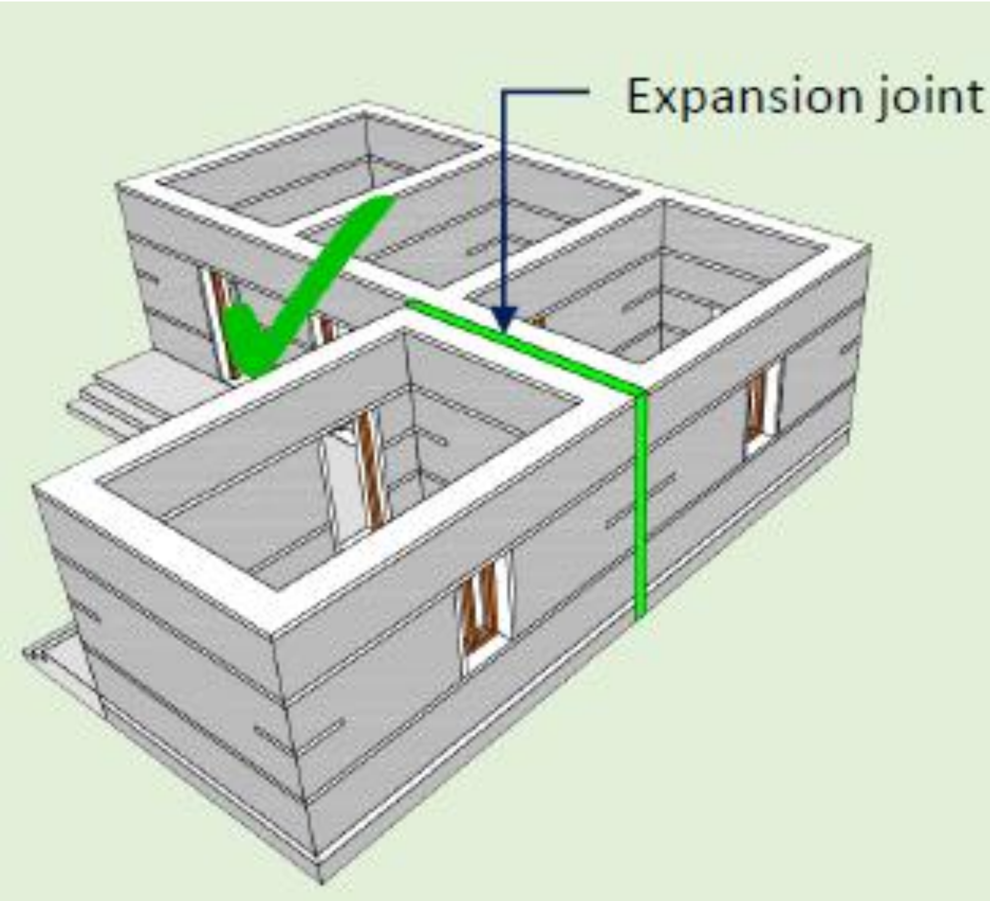
Adding Cross wall

2.2 Shape of Building Proportion

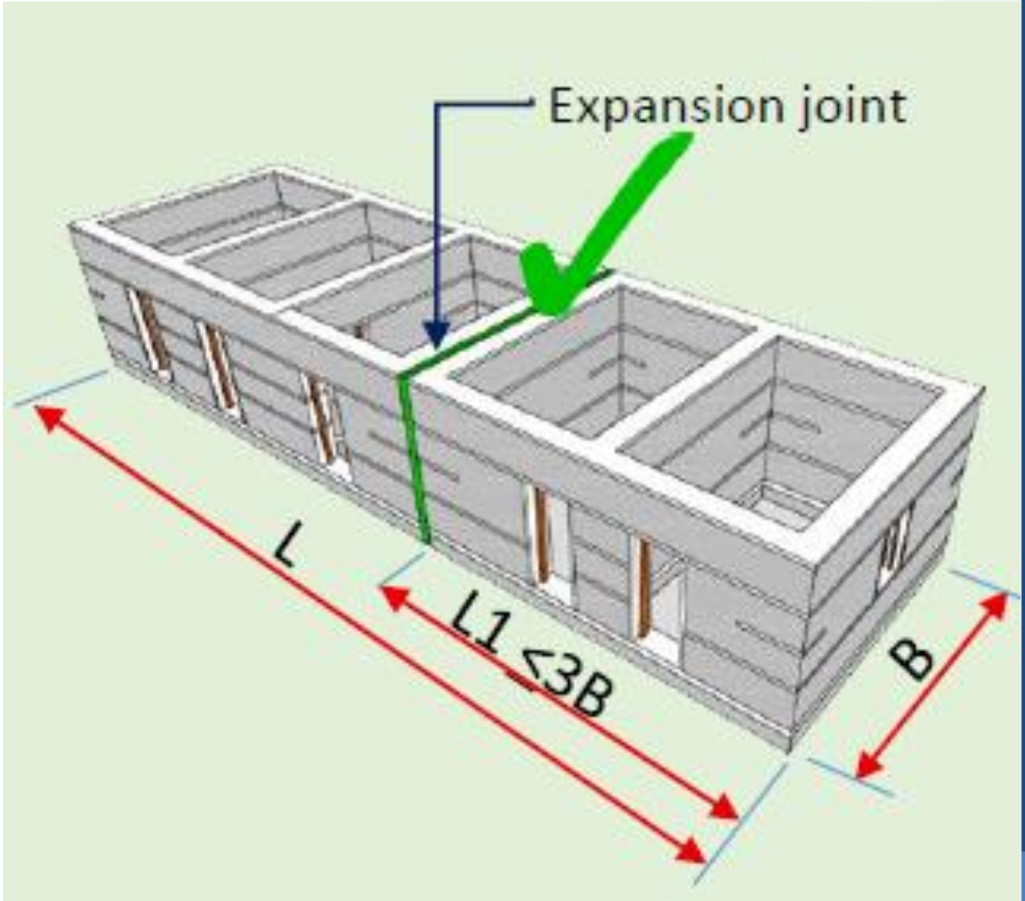


For small residential buildings not exceeding 100 sq. m. in plinth area with flexible floor and cross walls, the shape criterion of building can be ignored.

2.2 Shape of Building Proportion Contd.

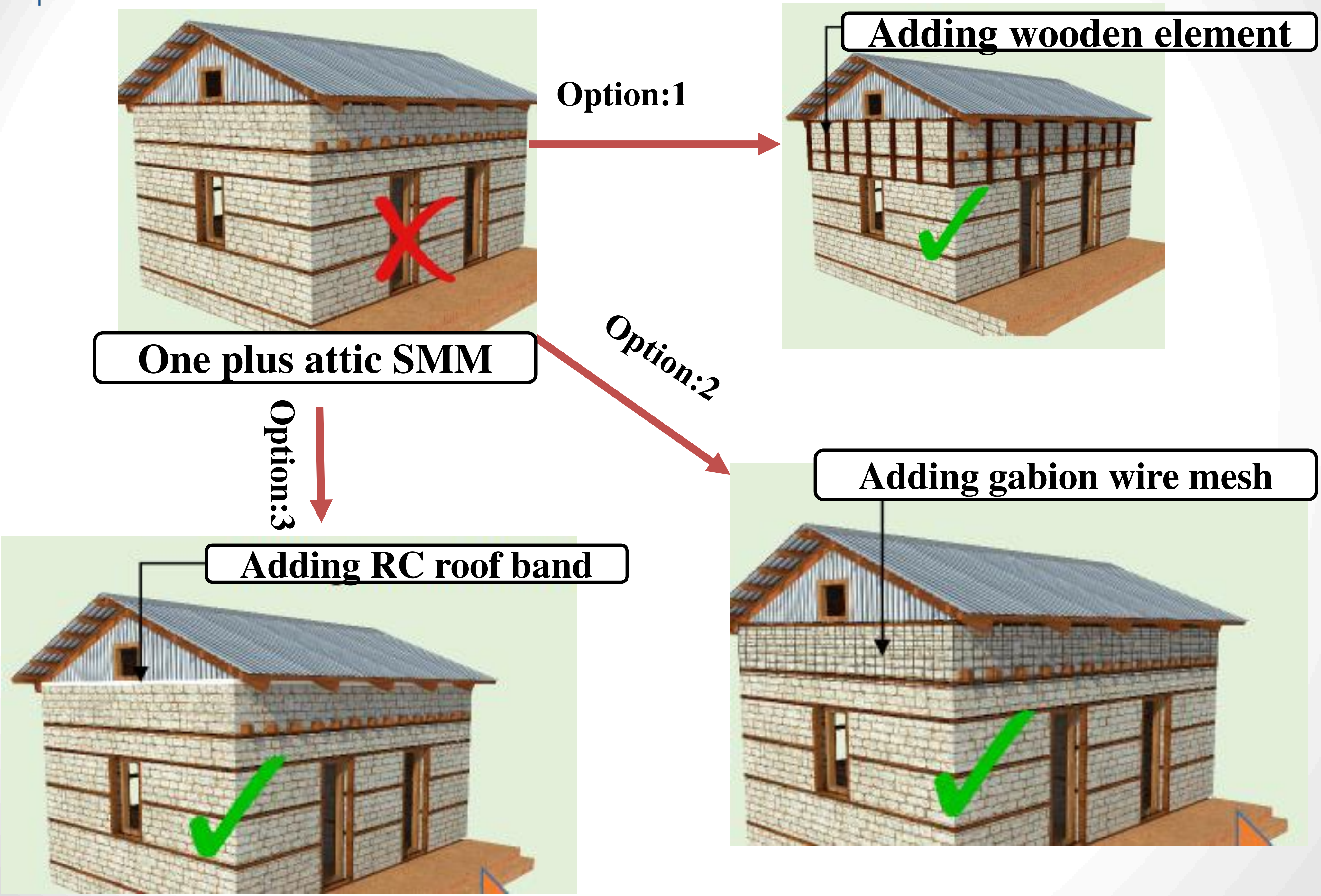


Expansion Joint for L-shape



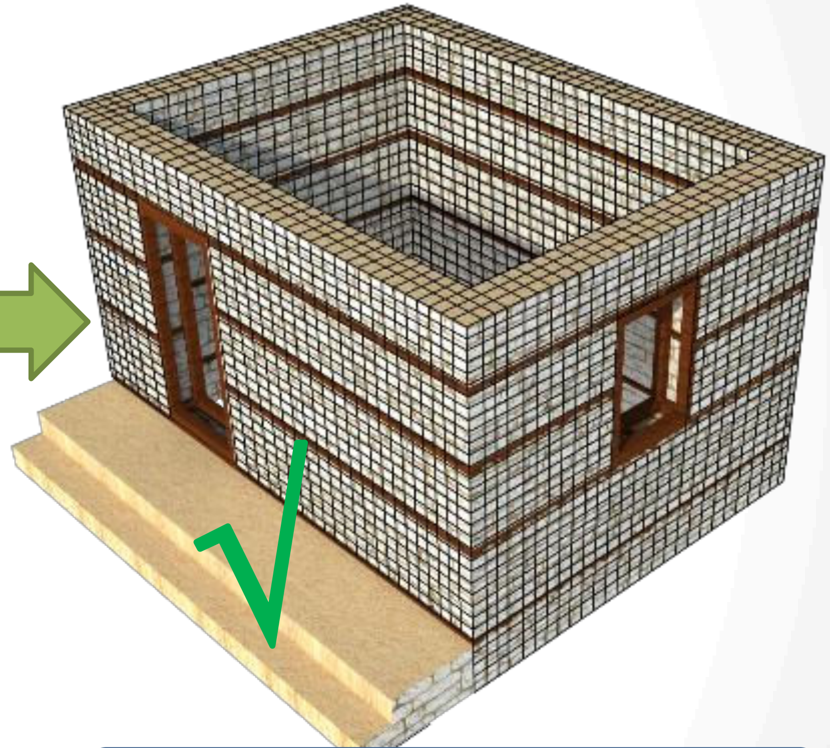
Expansion Joint for long building

2.3 Number of Storey



3. Using Improper Materials, Mix materials

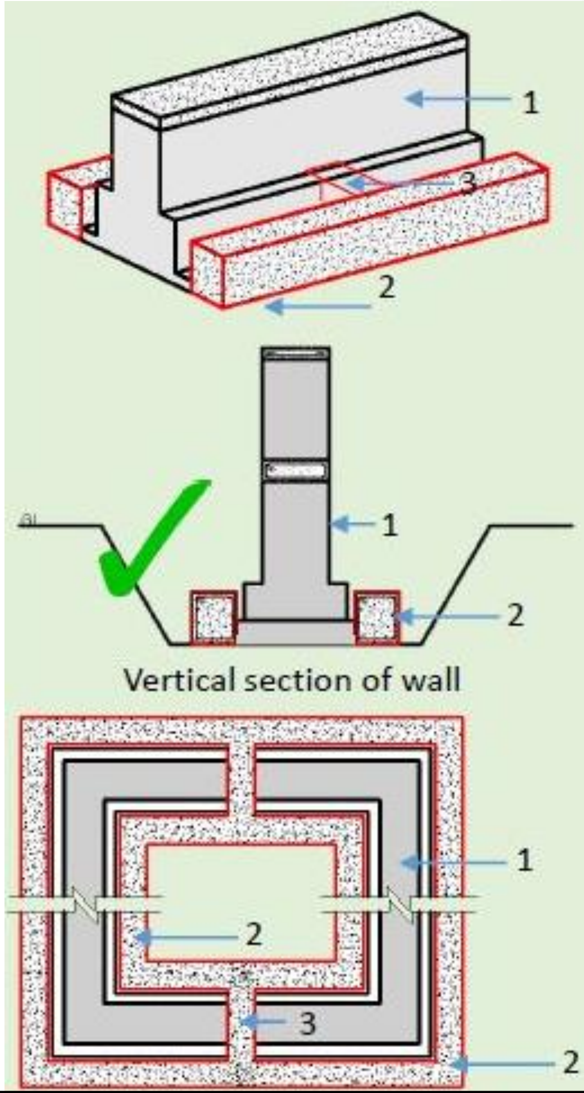
Mix use of stone and brick in the wall along the same direction.



Provide jacketing

Limitation: The masonry should have all the earthquake resistant features

4. Insufficient Foundation



- **Insufficient Foundation**
  - **Depth**
  - **Width**
  - **Shape**

Foundations size can be variable for rock bed.

1. **Foundation**
2. **Now concrete beams**
3. **Connecting lateral concrete beams**

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5.1 R.C Vertical Member

No RC Vertical Member

Option: 1

Option: 2

Provide reinforcement member at outside

Provide bandage at outside

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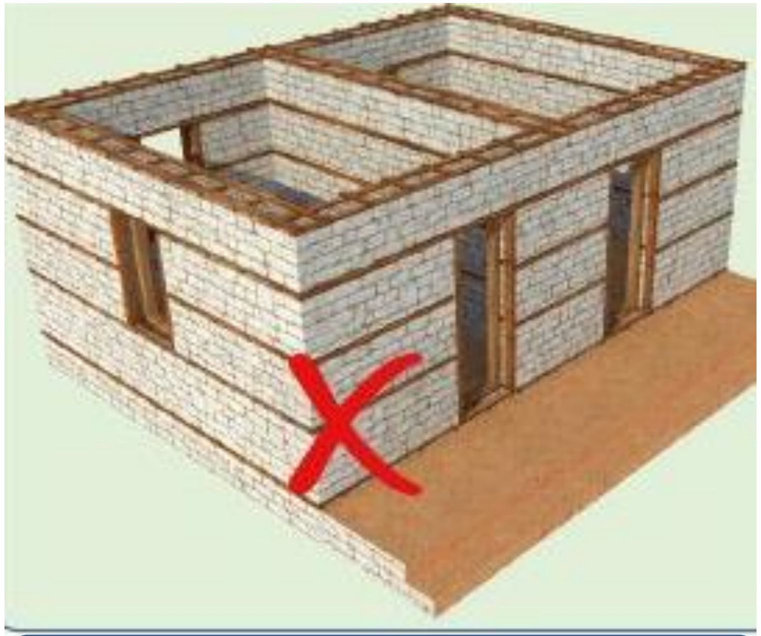
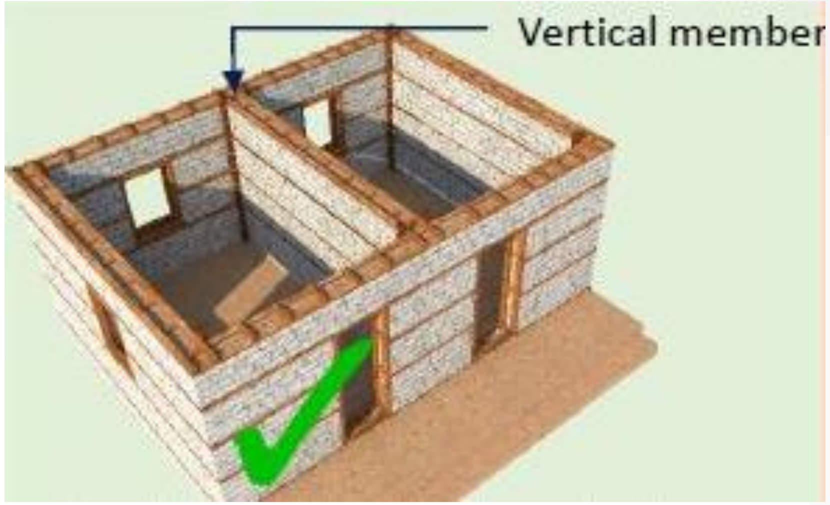
5.2 Wooden Vertical Member



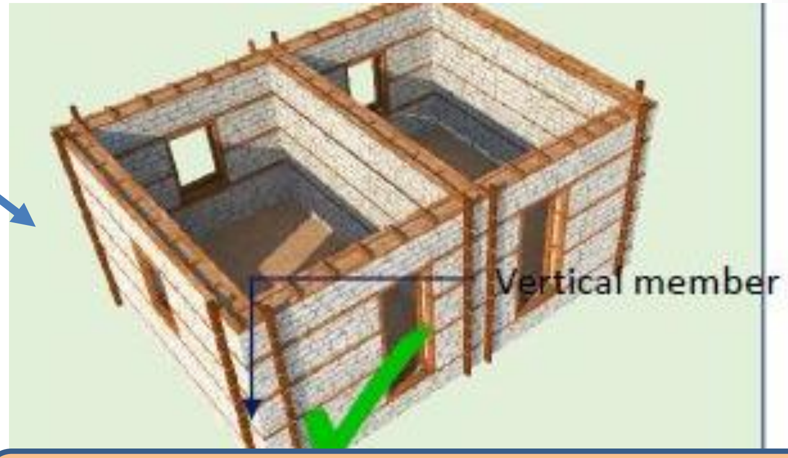
No vertical member

Option: 1



Installation of vertical member inside wall

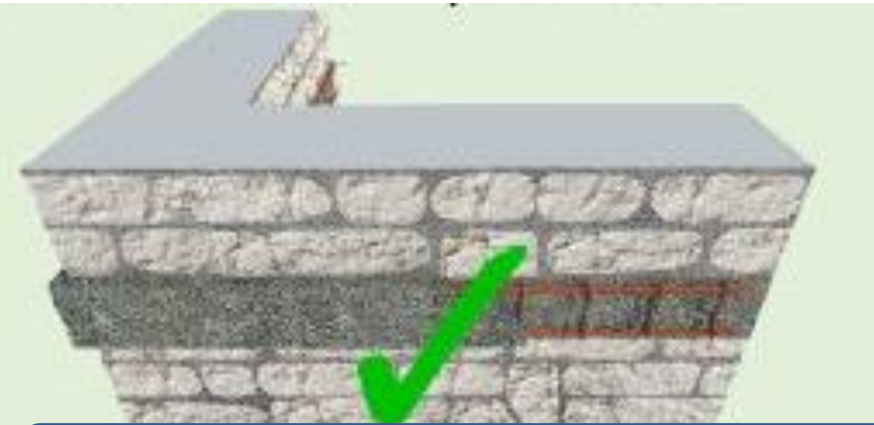
Option:2



Installation of vertical member inside wall

6.1 No Plinth beam/Level of Plinth

For no Plinth beam

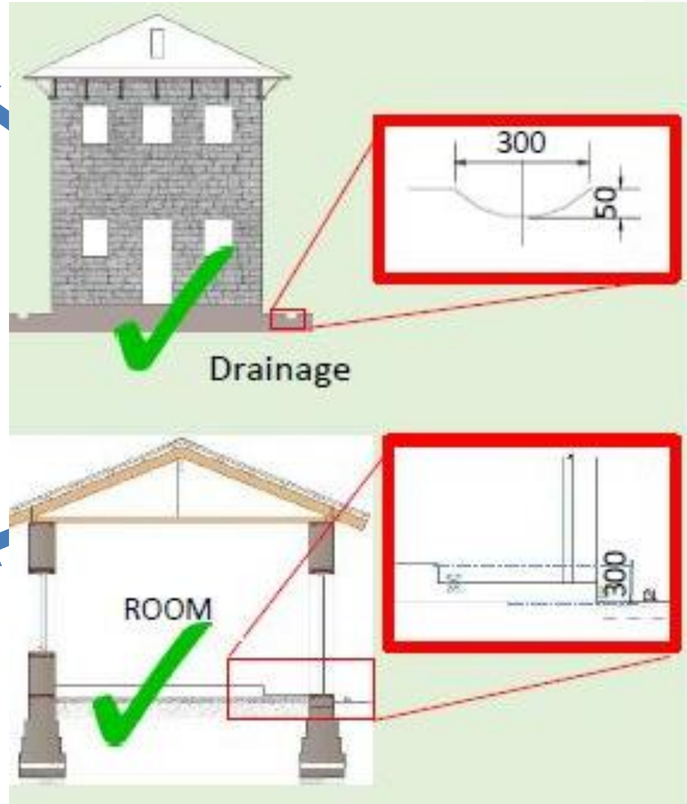


Provide external plinth band

For low level of plinth

1

2

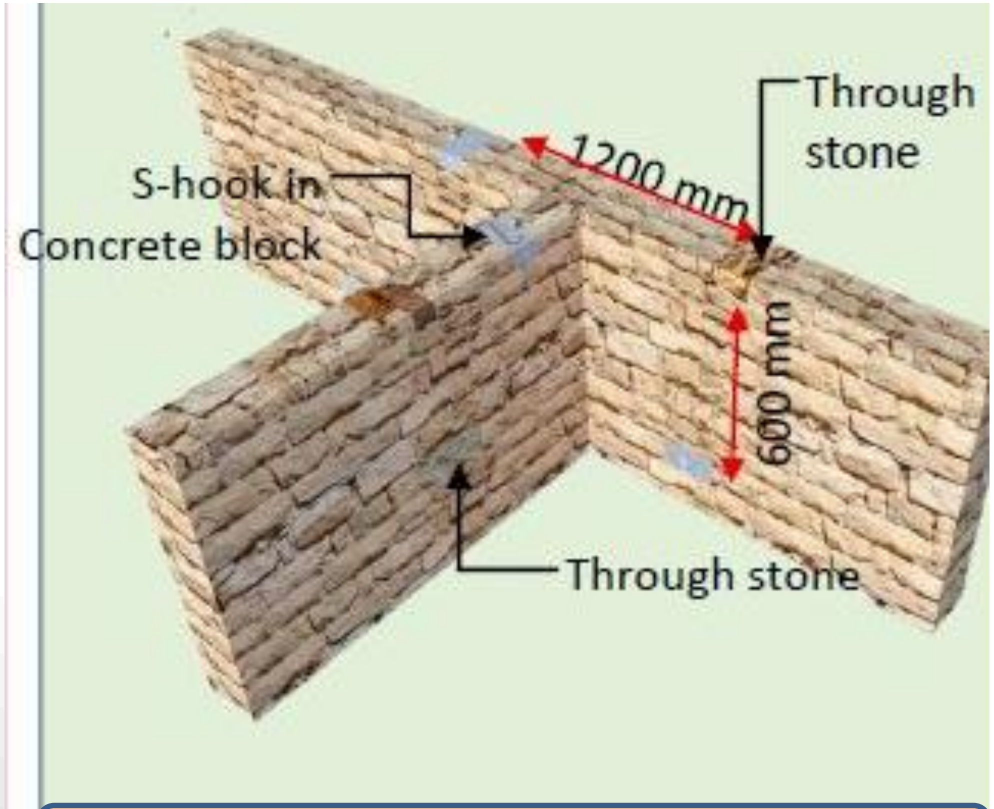


1. **Improving site drainage**
2. **Increasing floor level**

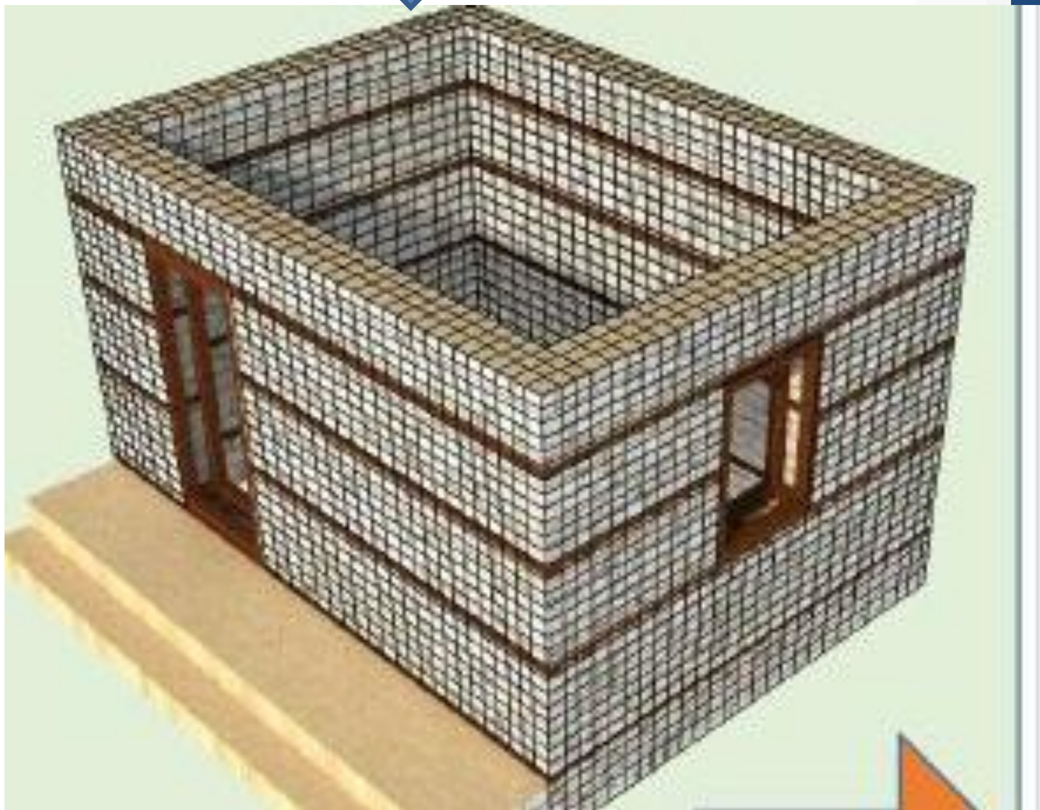
7.1 Weak Masonry/Lack Of through Stones

For absence of through stone

For weak masonry



Insert through stone

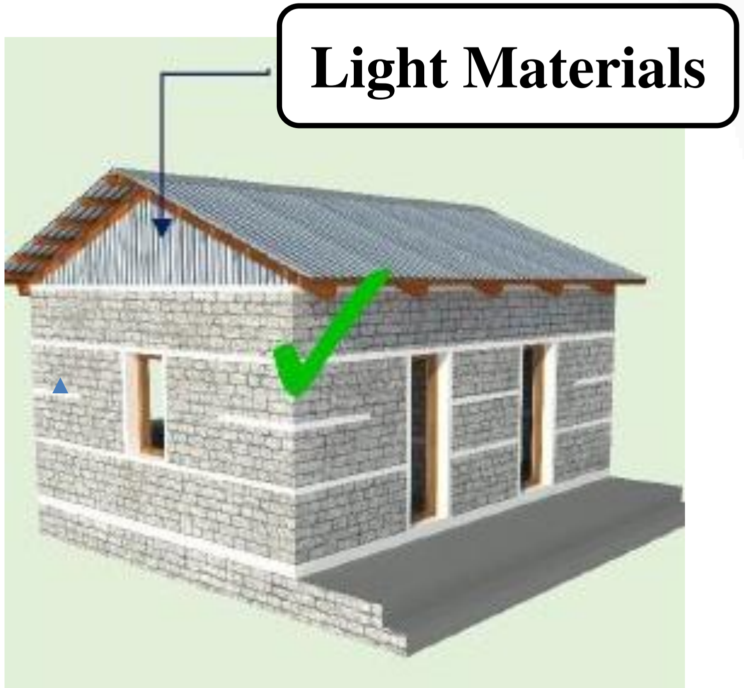


Jacketing for masonry wall

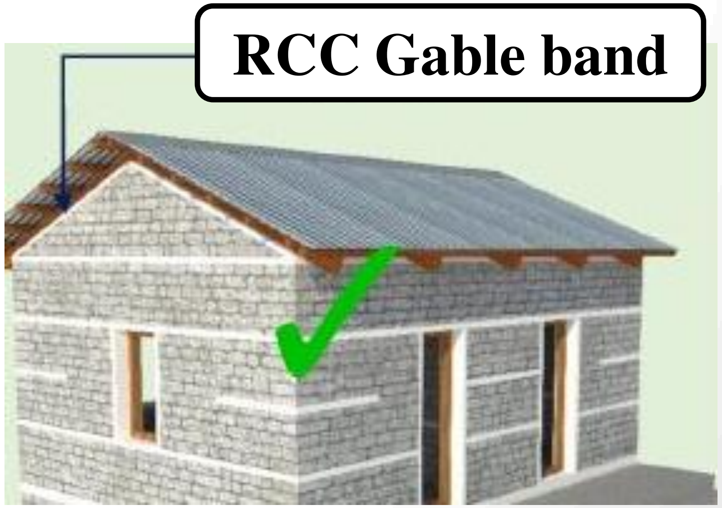
7.2 Vulnerable Gable wall

Unsupported gable wall

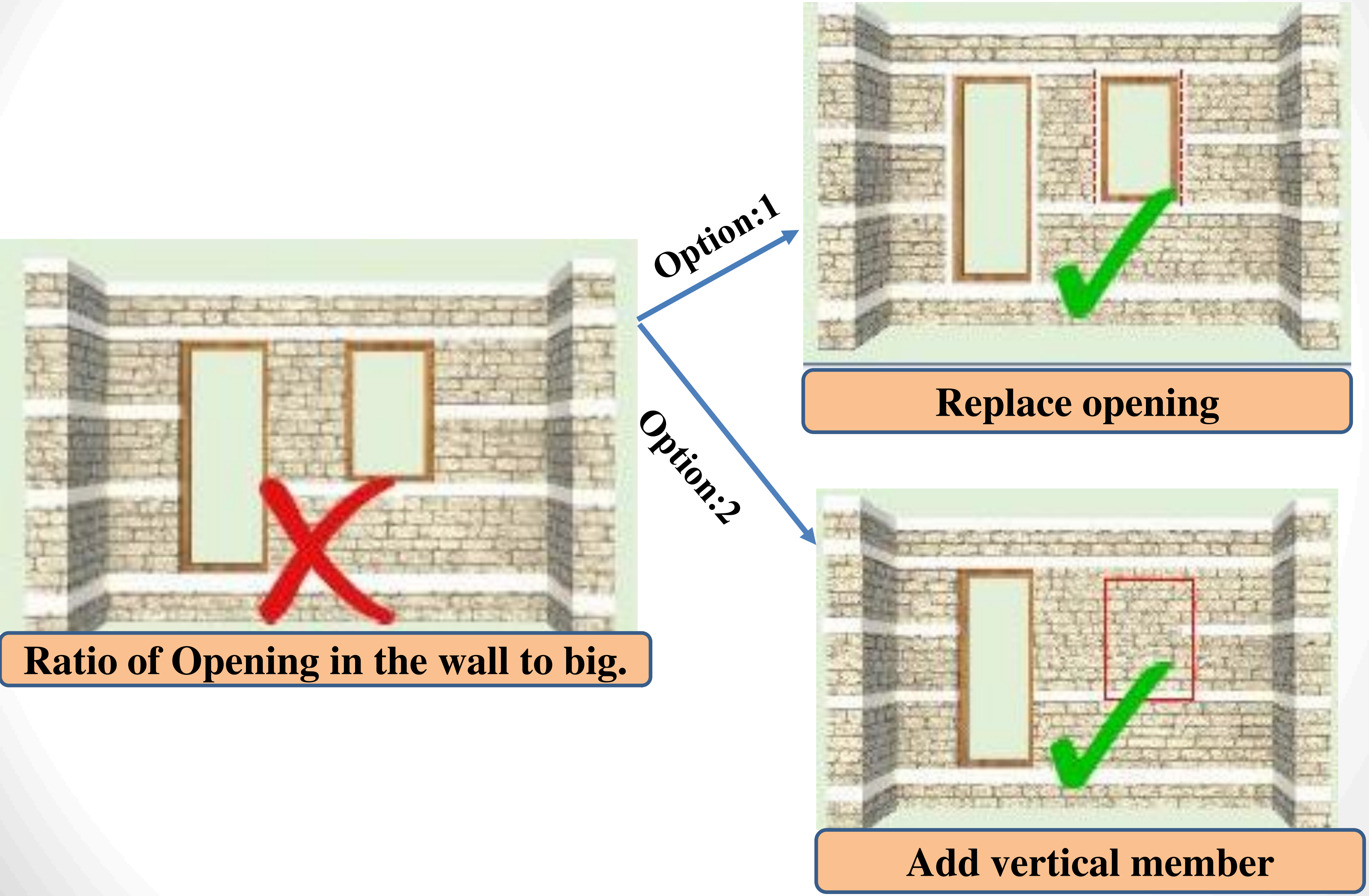
Option:1



Option:2



8.1 Inappropriate Position & Size of Opening



9.1 R.C Horizontal band

No horizontal band

Option: 1

External RC horizontal band

Option: 2

GI wire mesh bandages

9.2 Wooden Horizontal Band

No horizontal band

Installation of wooden external horizontal band

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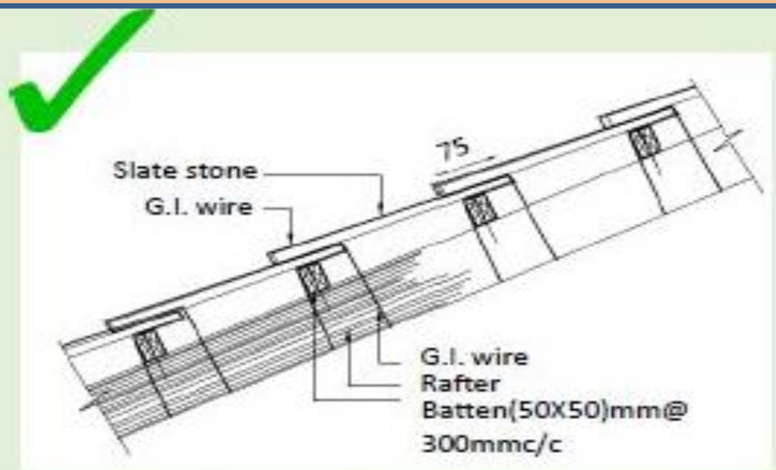
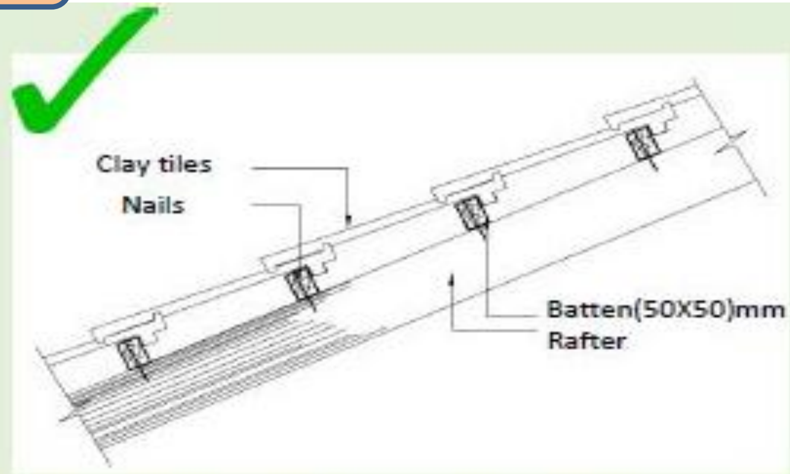
10. Connection/ Heavy material for Roofing

Heavy materials for roofing

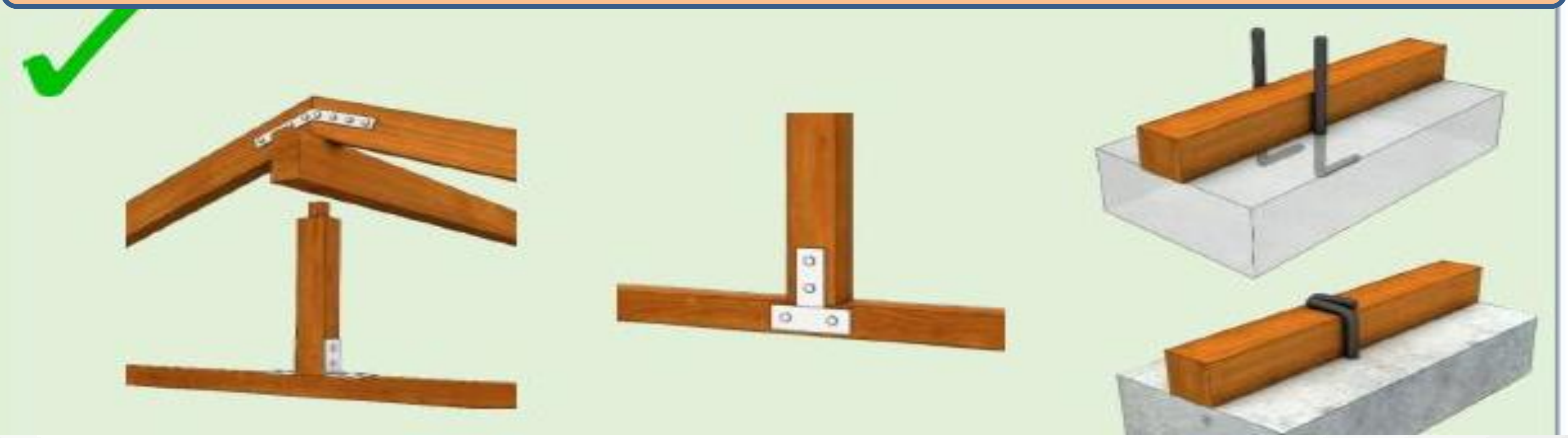


Detail of anchoring slate stone
Source: NBC203 2015, P44



Detail of fixing clay tile
Source: NBC203 2015, P45

Connections details of truss & band with metal strip



Part B: Mitigation Measures

- The mitigation measures consists of **step by step Procedure** for implementing the correction

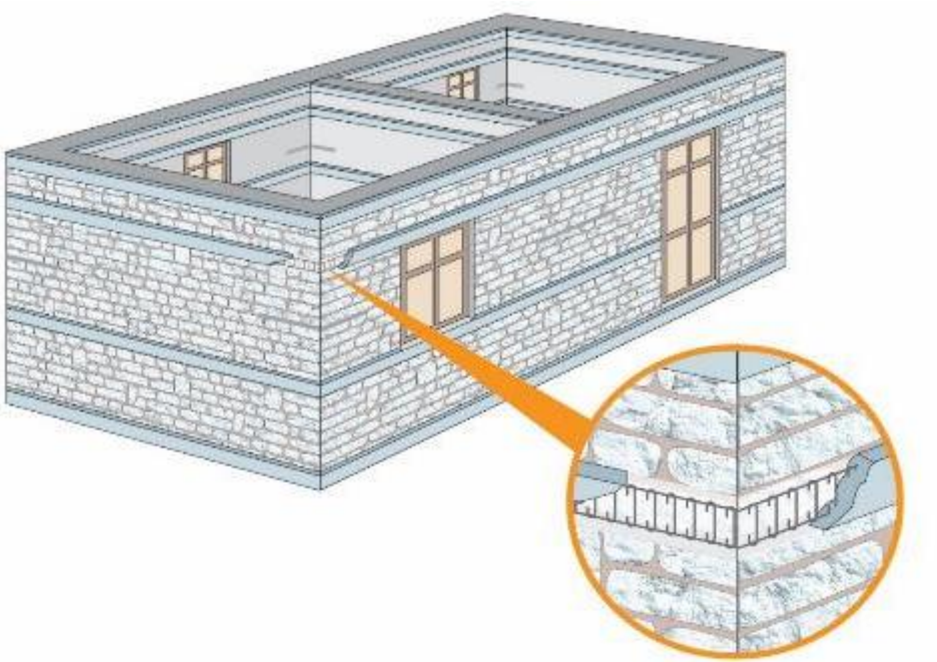
- Depending upon the availability of materials and workmanship suitable mitigation measures can be adopted

Included Components

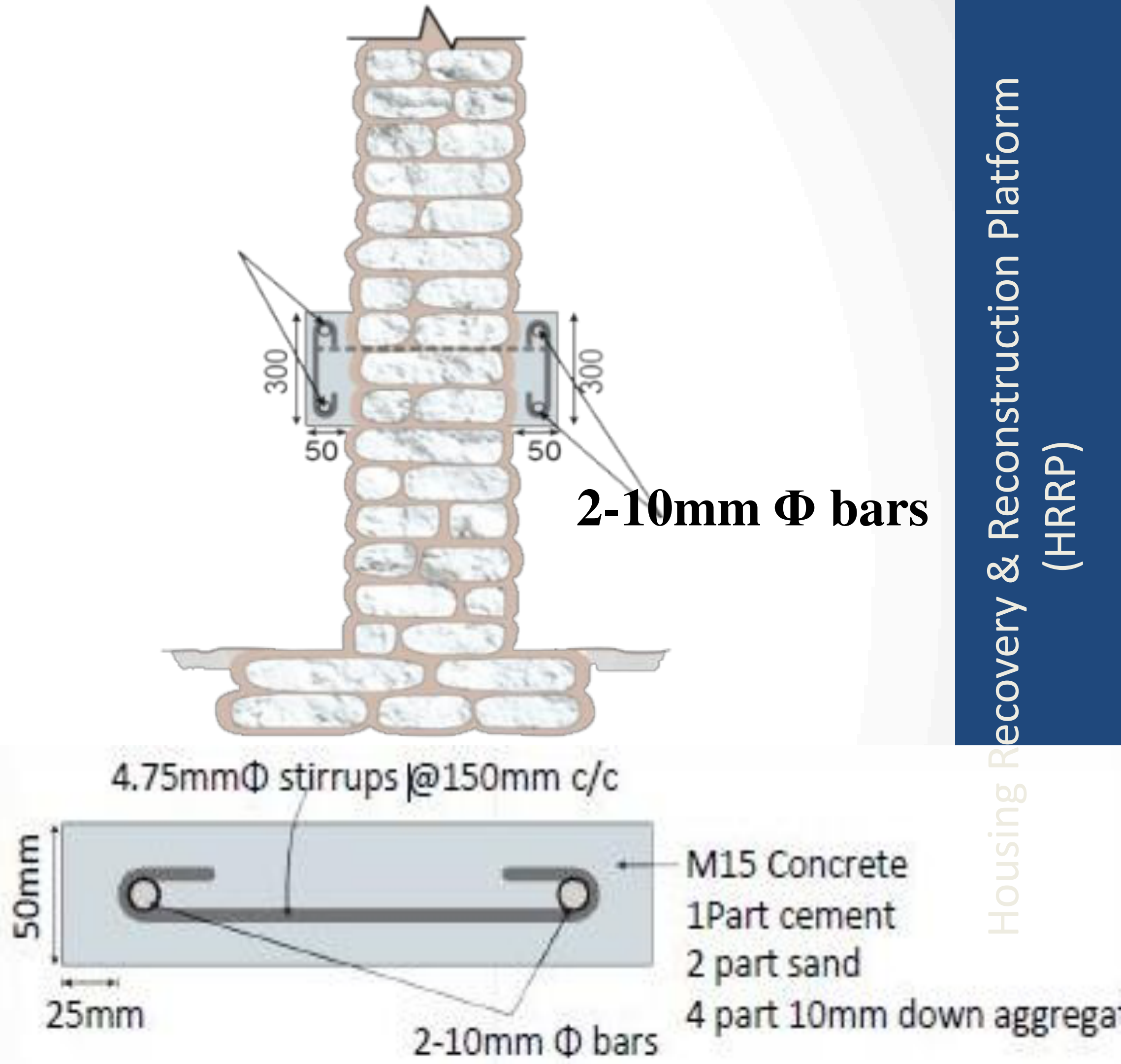
1. Constructing of retaining wall
2. Strengthening wall with Buttressing
3. Adding cross wall
4. Installation RC vertical reinforcement
5. Installation GI mesh for splint
6. Installation of wooden vertical member
7. Strength opening
8. Adding RC horizontal band
9. Adding bandage
10. Adding horizontal band
11. Strength wall by jacketing

Examples

Installation of RC Horizontal Reinforcement

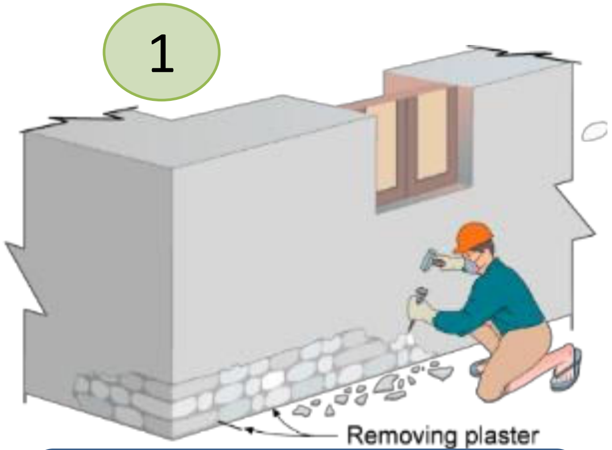


Addition of bandages on outer faces of wall when plinth/ sill/ lintel/ roof bands are missing

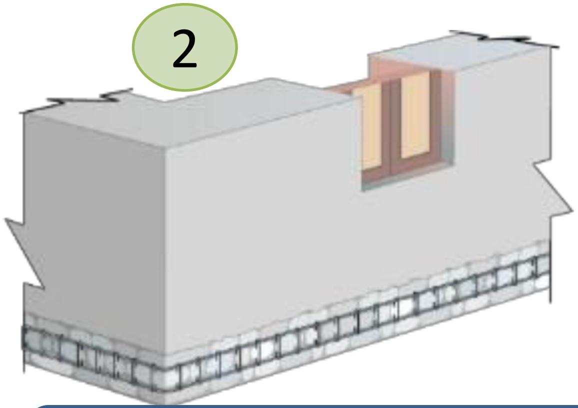


Details of Splints

Construction Procedure
(Adding RC Horizontal band)

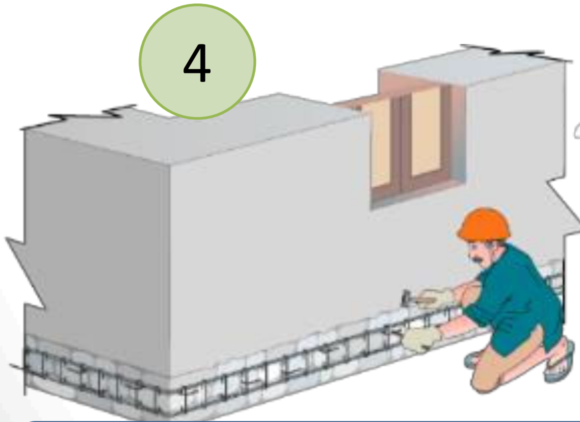


Surface Preparation



Placing of reinforcement

Making holes of anchorage



Anchor reinforcing bar mesh

Application of Micro Concrete

Curing of Concrete

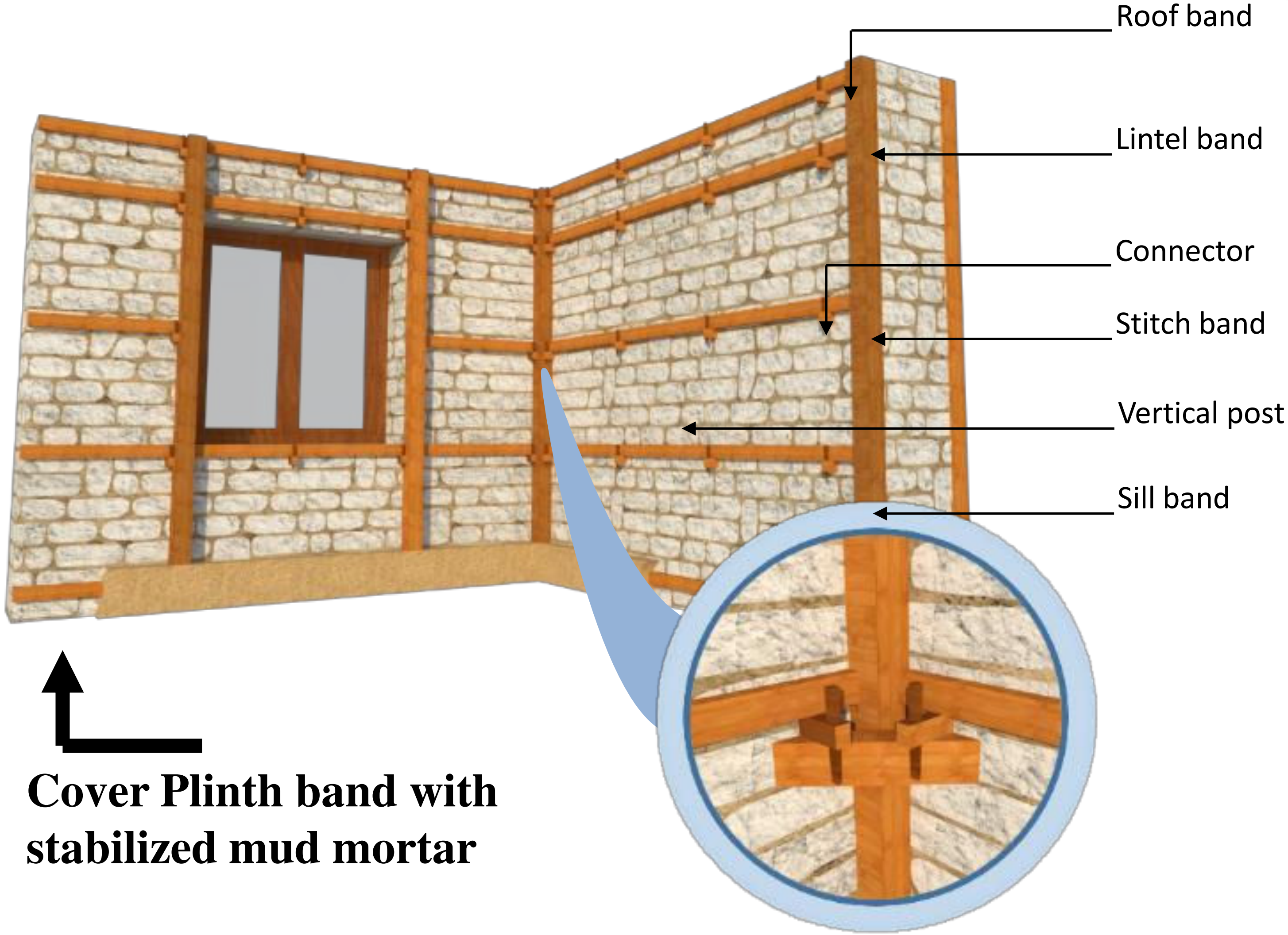
Annex

- **It Covers details of corrective measures.**
- **It provides different options with detailing to place Earthquake Resistant Element which is included in the corrective measures guide.**

Annex I: Options of Corrective measures

1. **Wooden Frame with Wooden Connection**
2. **Wooden Frame and Corner with Diagonal Bracing with Wooden Connection.**
3. **Wooden Frame with iron nut bolt Connection**
4. **Wooden Frame & Corner Diagonals with iron rod Connection**
5. **Gabion Wire Wrappings**
6. **Precast Concrete Frame with Iron rod Connection**

Wooden Frame with Wooden Connection (For Example)



Cover Plinth band with stabilized mud mortar

Annex II: Options of Corrective measures

1. Use of Wooden/ Bamboo sections
2. Use of Wooden / Bamboo Sections (Both Side)
3. Use of Iron Sections
4. Use of Iron Sections with Prefab Panels

Use of Wooden/ Bamboo Sections (For Examples)

Annex II Provides details of Hybrid structures only

Use of Manual

Beneficiary

Apply for Inspection

→

GP/NP

Receive &Mobilize IT

If ok →

Inspection Team

Approve and recommend for next tranche using Form Annex:10

If not →

Inspection Team

Suggest suitable Corrective measures as Specified in Manual using form Annex:11

The same process repeat after the beneficiary do the corrective measures as instructed by inspection team

Updated Inspection Forms

Lists of Updated forms

- Stone in Mud_ First Inspection
- Stone in Cement _First Inspection
- Brick in Mud _First Inspection
- Brick in Cemet_First Inspection
- Stone in Mud_Second Inspection
- Stone in Cement_Second Inspection
- Brick in Mud_Second Inspection
- Brick in Cement_Second Inspection
- RCC _First Inspection
- RCC_Second Inspection
- Confined Masonry_First Inspection
- Confined Masonry_Second Inspection

Corrective measures incorporated in Inspection Forms

प्राविधिक सहायक	□छ □छैन	संस्था	□नेपाल सरकार□गैरसरकारी संस्था१ ०
तालिम प्राप्त डकर्मी प्रयोग गरिएको	□छ □छैन	माटोको प्रकार	□कडा□मध्यम□नरम

खण्ड -२ : विस्तृत प्राविधिक विवरण

नं	वर्ग	विवरण			(Correction/exception manual अनुसार)	न्यूनतम मापदण्ड पालना गरिएको		टिप्पणी
						छ	छैन	
१	निर्माण स्थलको छनोट निम्न स्थान बाट टाढा हुनुपर्छ।	भौगर्भिक चिरा परेको ठाउँ।			(बस्ति स्तान्तरण सिफारिस गरिएका बाहेक अन्य क्षेत्रमा आवासिय प्रयोजनका घरको लागी मुल्यांकन गर्न नपर्ने।)	□	□	
		भिरालो क्षेत्र > २०°।			(refer details of retaining walls given in Correction/exception manual)	□	□	
		पहिरो जान सक्ने क्षेत्र।				□	□	
		नदीको बगर वा सिमसार ठाउँ।			(उपलब्ध जमिनलाई निर्माण उपर्युक्त बनाइ निर्माण गर्न सकिने। बाढीको अधिक्तम सतह र नदी किनारबाट न्युनतम दुरी (सरकारी/नगर/गाउँपालिका मापडण्डमा उल्लेख भए बमोजिम)मा निर्माण गर्न सकिने।)	□	□	
		ढुङ्गा झर्ने ठाउँ।			(बस्ति स्तान्तरण सिफारिस गरिएका स्थान, भौगर्भिक अध्ययन तथा स्थानिय चलन चल्तिले वर्जित स्थान बाहेक अन्य क्षेत्रमा निर्माण गर्न सकिने।)	□	□	
		तरलीकरण हुन सक्ने ठाउँ।				□	□	
		माटो भरेको वा पुरुवा माटो भएको स्थान।			(घरको जगलाई पुरीएको माटो भन्दा तल राखी निर्माण कार्य गर्न सकिने।)	□	□	
२	भवनको आकार प्रकार र नाप	तल्ला संख्या	प्रचलित ढलान पट्टी	एक तल्ला र बुईगल सम्म सिमित	Refer correction exception manual) (घर संरचनात्मक हिसावले सुरक्षित भएको हुनुपर्ने र संरचनात्मक सुरक्षा मुल्याङ्कन सहित पेश गर्नुपर्ने।)	□	□	
			काठको प्रचलित पट्टी	एक तल्ला सम्म सिमित				

Housing Recovery & Reconstruction Platform (HRRP)

New Approved Housing Designs

Lists of New Housing Designs

- Design Catalogue Volume-II
- RCC Type Designs
- Load Bearing Type Designs
- Single Storey RCC Type Designs
- Dry Stone Masonry Type Designs

Catalogue for Alternative Construction Materials and Technologies

DESIGN CATALOGUE FOR RECONSTRUCTION OF EARTHQUAKE RESISTANT HOUSES

VOLUME II

MARCH, 2017 (FALGUN,2073)

GOVERNMENT OF NEPAL
MINISTRY OF URBAN DEVELOPMENT
DEPARTMENT OF URBAN DEVELOPMENT AND BUILDING CONSTRUCTION
BABARMAHAL, KATHMANDU

Housing Recovery & Reconstruction Platform (HRRP)

Introduction

- **DUDBC has prepared the document.**
- **It covers seventeen model design following twelve technologies.**
- **The designs provided in this catalogue are based on calculations, model test and analytical tests as these technologies are not covered by Nepal National Building Code, 2060.**
- **It covers Alternative Construction Materials and Technologies.**

Catalogue Objective

To pave way for use of alternative materials and technologies in the reconstruction process as per the principles set by Post Disaster Needs Assessment (PDNA) for housing and human settlements recovery and reconstruction.

Catalogue Limitation

- **Designs included in this catalogue can be selected and used as they are, for reconstruction of urban and rural housing .**

- **Designs, other than those included in this catalogue, detailed engineering design and approval from concerned authorities shall be done.**

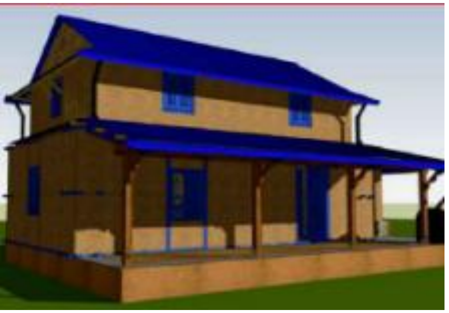
Lists of Model

1. **Interlocking Brick Masonry One Storey**
   - One Storey
   - One Storey
   - Two Storey
2. **Confined Hollow Concrete Block Masonry**
   - Two Storey
3. **Hollow Concrete Block Masonry**
   - Two Storey
4. **Compressed Stabilized Earth Block Masonry**
   - One Storey
   - Two Storey
5. **Random Rubble Masonry in Mud mortar with GI wire Containment**
   - One Storey
   - Two Storey
6. **Bamboo and Stone Masonry Hybrid Structure**
   - Two Storey
7. **Rat Trap Bond Masonry**
   - One Storey
8. **Earth Bag Masonry**
   - One Storey
9. **Light Gauge Steel Structure**
   - One Storey
   - Two Storey
10. **Steel Structure**
    - Two Storey
11. **Timber Structure**
    - Two Storey
12. **Debris Block Masonry**
    - Two Storey

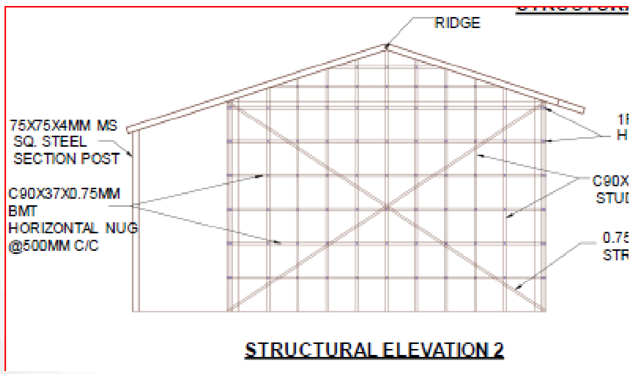
Alternative Construction Materials












RCC Type Designs
(Non Compliant Structures)

Lists of Type Designs

- 1 bay x 4 bay Building
- 1 bay x 3 bay Building
- 1 bay x 2 bay Building
- Building having Cantilever projection 1.5 m
- Building having Grid spacing 1.5 m c/c
- Building with Four Number of Columns, all eccentric
- Building having foundations at differential level
- Building having column size 9” x 12”
- Building with a Missing Column
- Building having Grid Size 4.5m x4.5m
- Building having Grid size 3.5mx 5.5m

Limitation

- **Can reflect only these design & drawing at field**

- **Detail structural analysis required for the building, if the dimension of the building differ from these designs.**

Load Bearing Type Designs (Low Cost Housing)

Housing Recovery & Reconstruction Platform (HRRP)

Brief

Four numbers of Stone Masonry Residential Building is approved by CLPIU, designed by SONA

- **Stone Masonry in Mud Mortar Single Storied -2 designs (wooden bands and vertical post)**
- **Stone Masonry in Cement Mortar- 1 Story (RCC band and rebars)**
- **Stone Masonry in Cement Mortar- 2 story (RCC Band and rebars)**

- **Stone Masonry in Mud Mortar Single Storied Building Submitted by JICA (wooden band and Vertical Posts)**

Dry Stone Masonry Type Designs

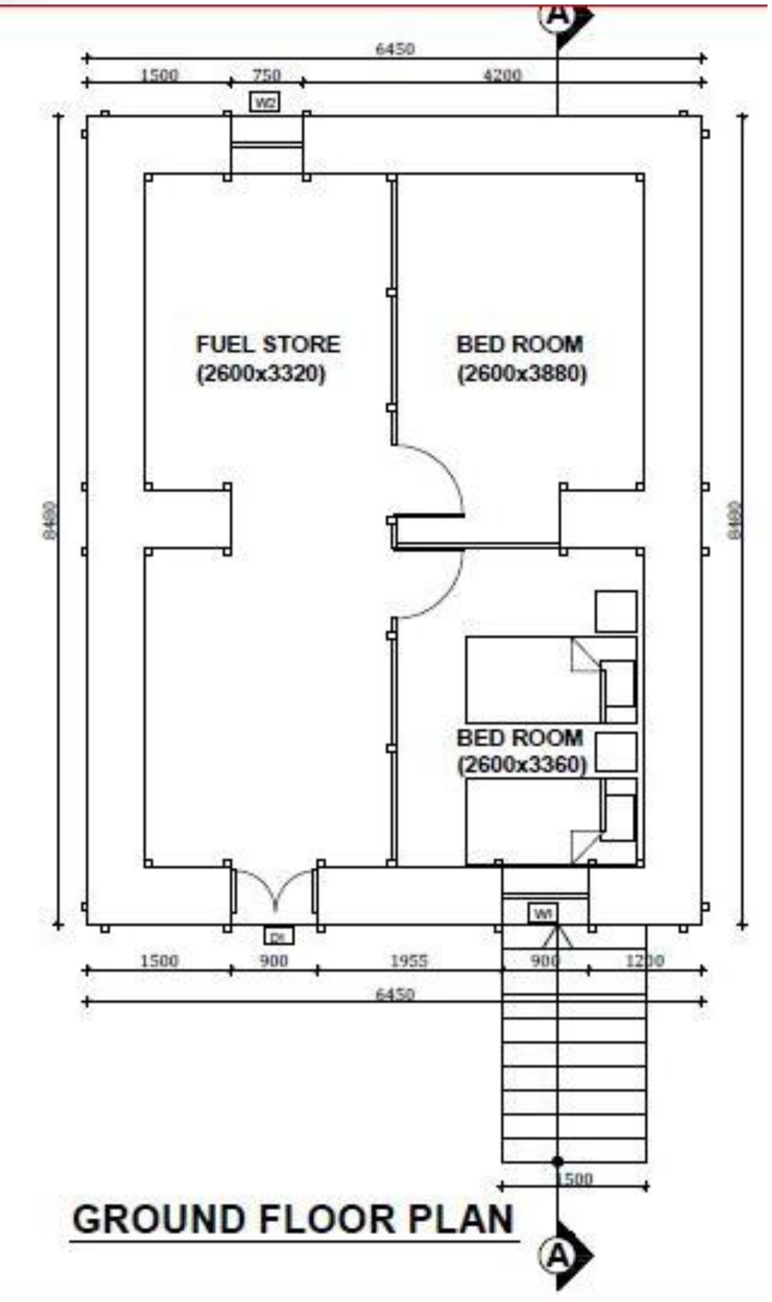
Types

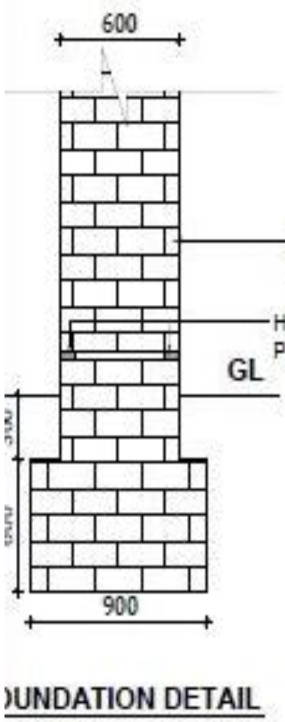
- **Dressed Stone Masonry**
- **Random Rubble Masonry**

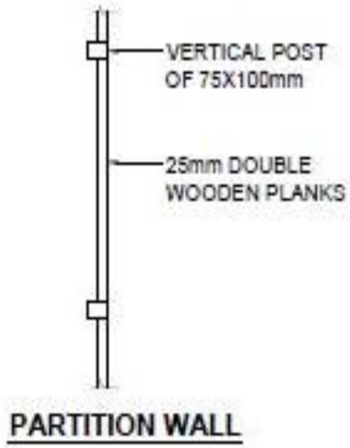
Dressed Stone Masonry

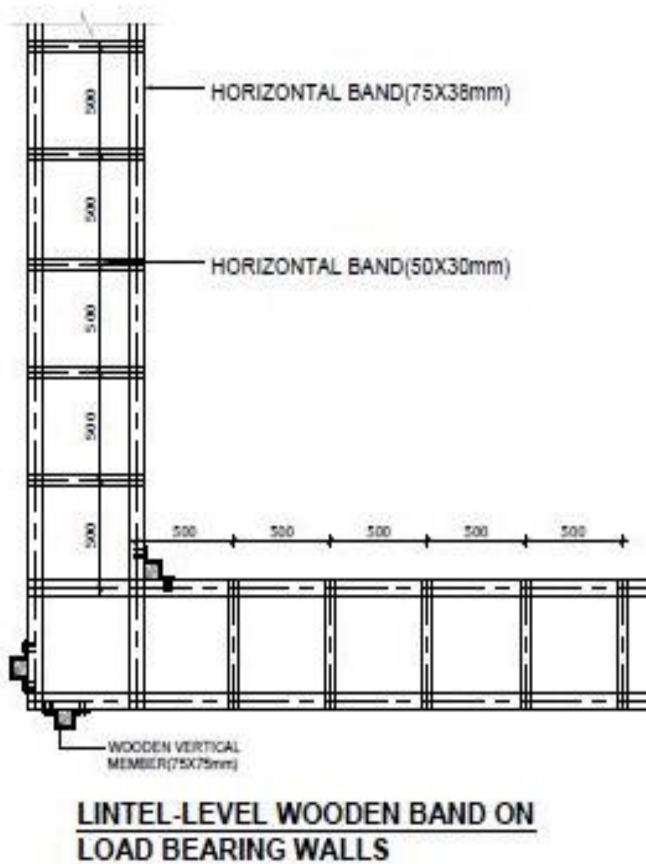
- Designed by MOUD/CLPIU
- Dry Dressed Stone Masonry Wall
- Suitable for Northern part
- Can reflect this technology for other building with different size

Dressed Stone Masonry





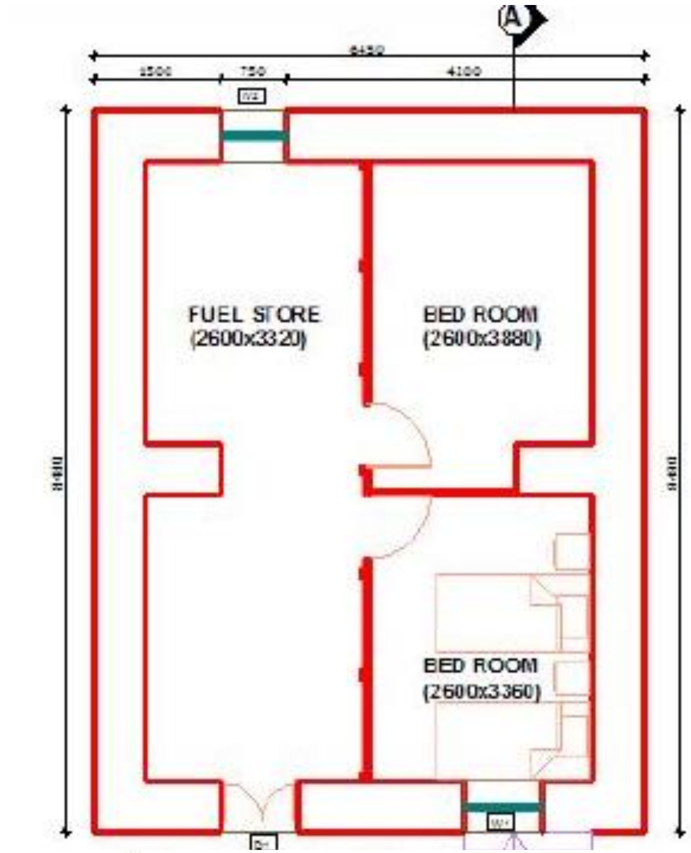


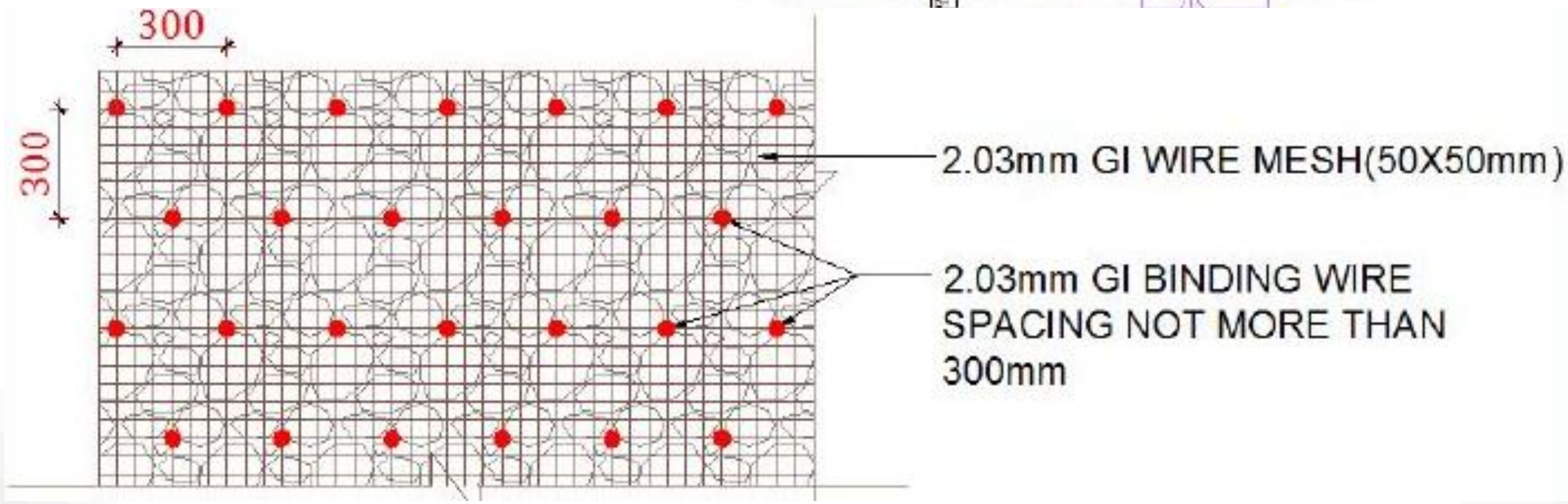


Random Rubble Masonry

- Designed by MOUD/CLPIU
- Dry Masonry wall
- GI wire mesh used in wall for Jacketing.
- Rubble stone used as a wall unit.

Random Rubble Masonry



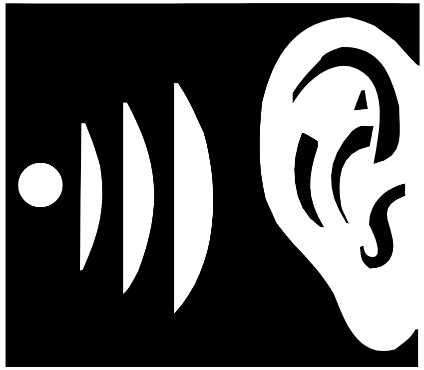


Housing Recovery & Reconstruction Platform (HRRP)

Communications Skills & FAQs

The Communication Facts

- What you <u>hear</u>
  - Tone of voice
  - Vocal clarity
  - Verbal expressiveness

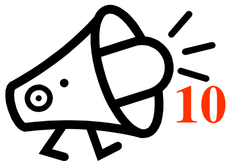


40% of the message

- What you <u>see or feel</u>
  - Facial expression
  - Dress and grooming
  - Posture
  - Eye contact
  - Touch
    message
  - Gesture

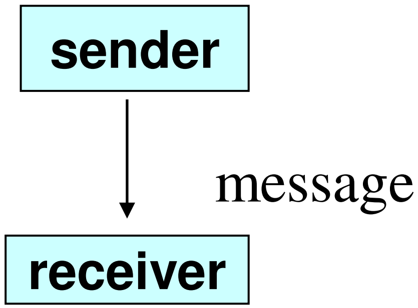
50% of the

- <u>WORDS</u> …



10% of the message!

Communication is a 2-way process

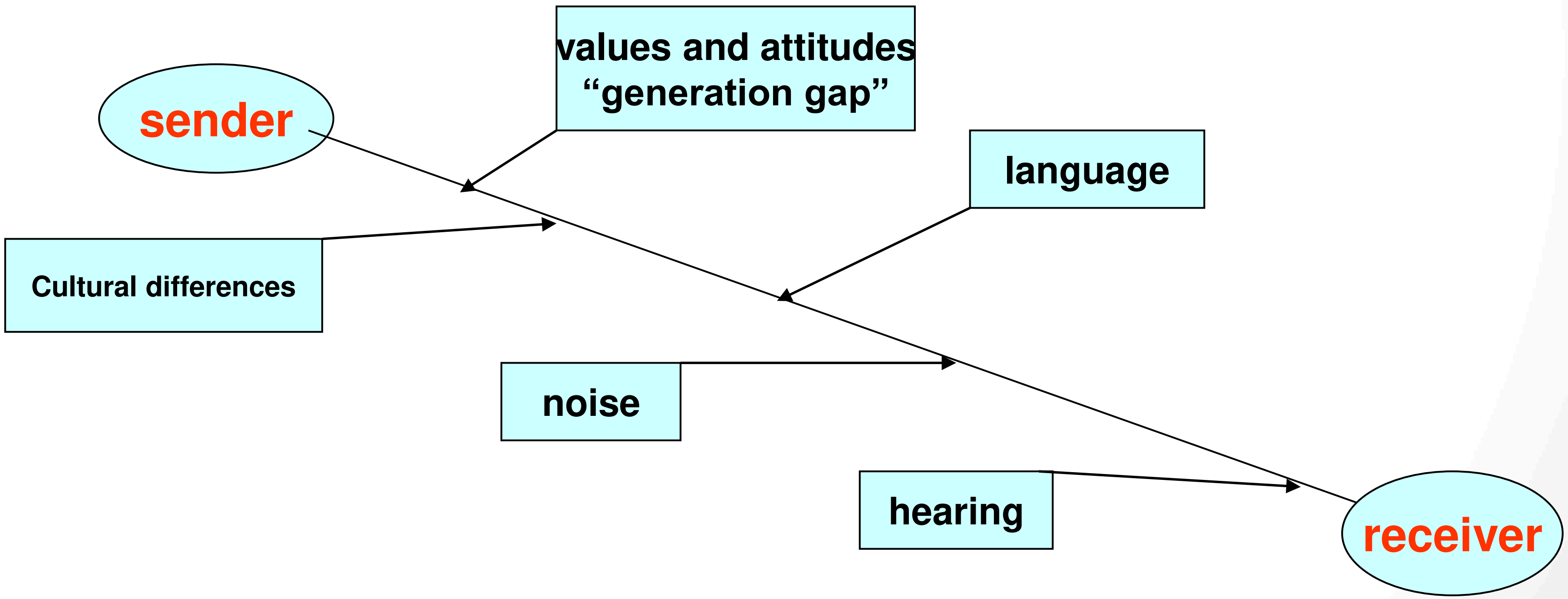


Communication skills involve:

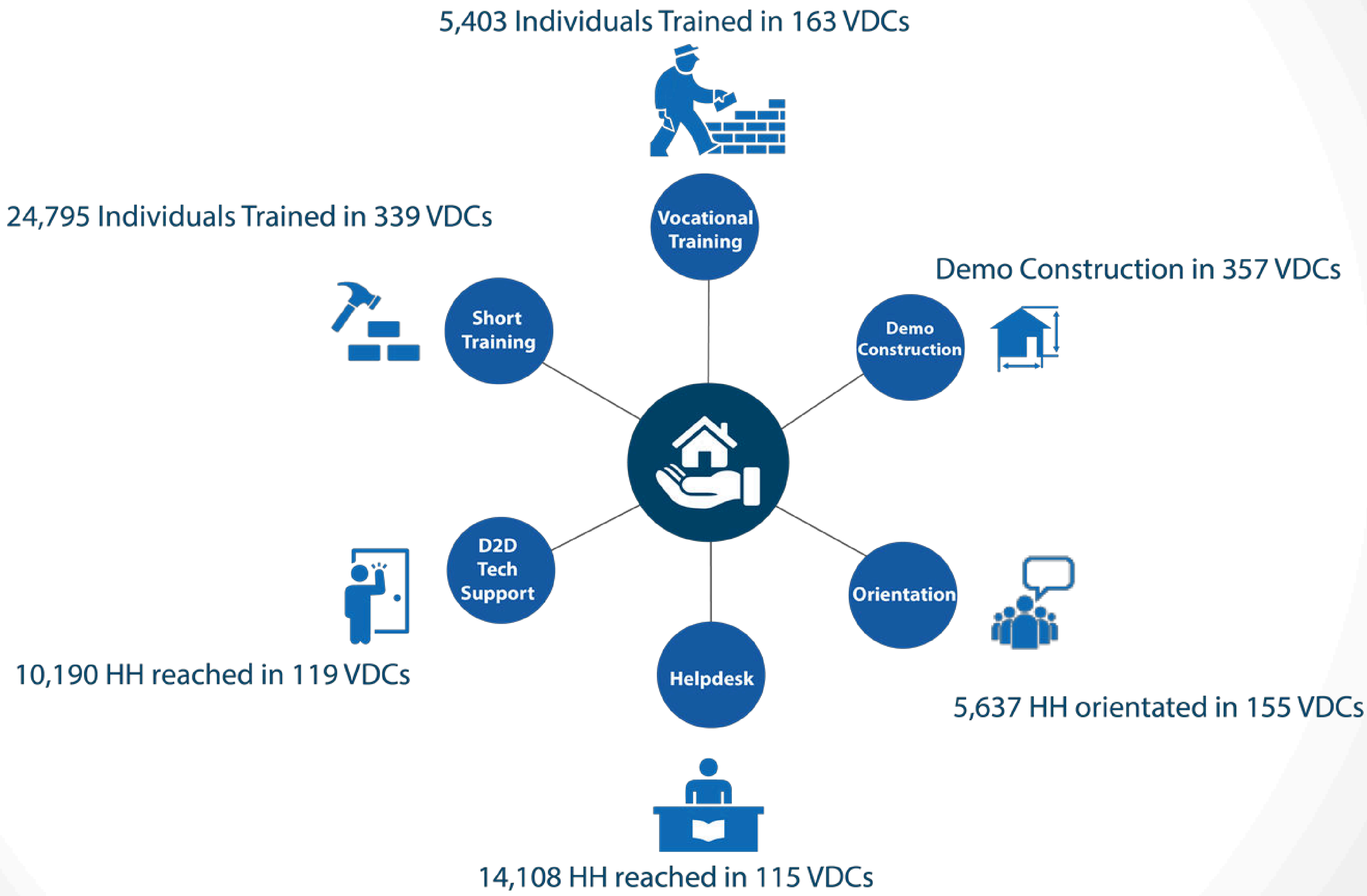
Listening to others (Receiving)

Asserting/ Expressing (Sending)

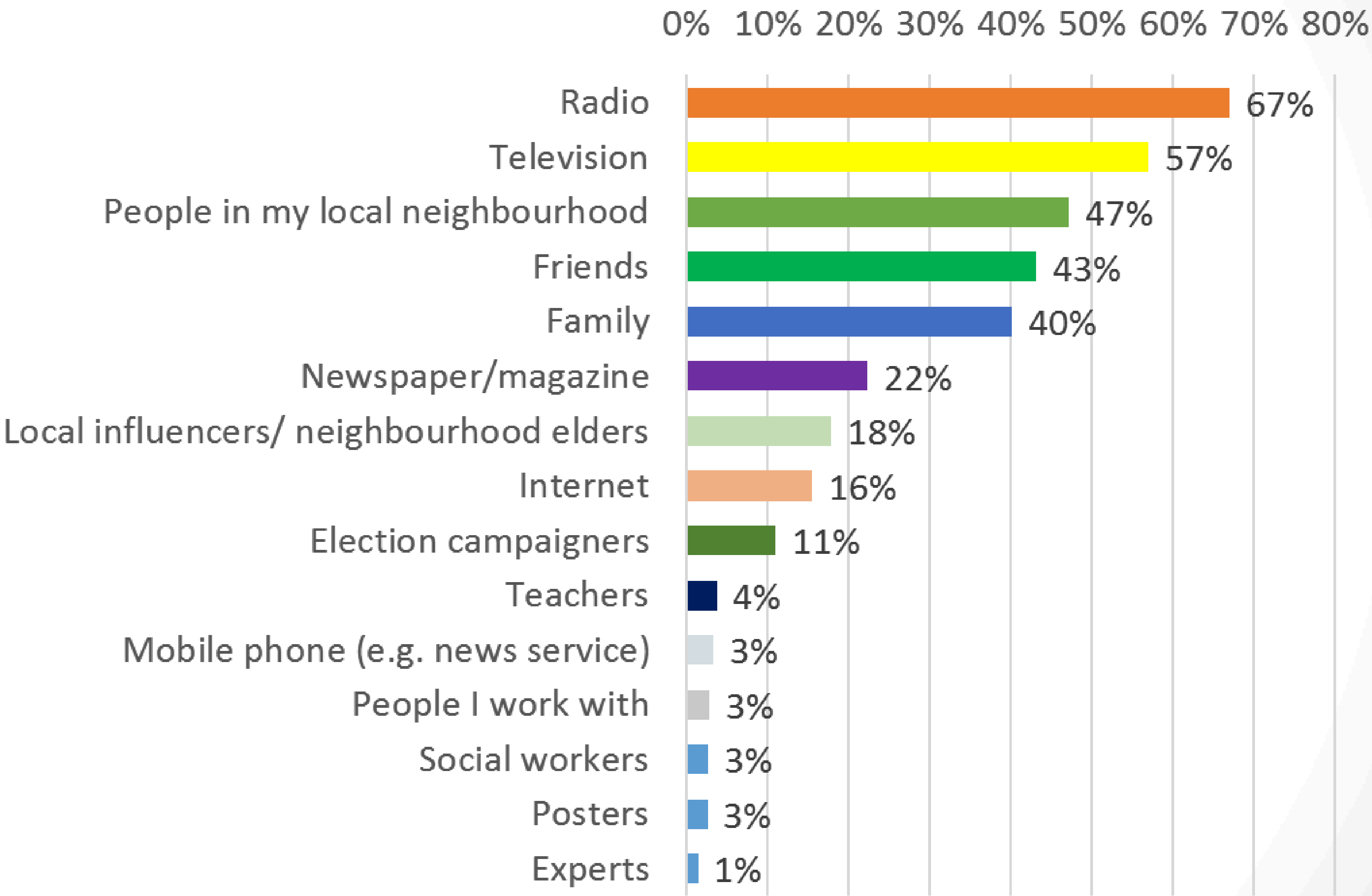
Barriers to communication can lead to misunderstanding and confusion



Housing Recovery & Reconstruction Platform (HRRP)

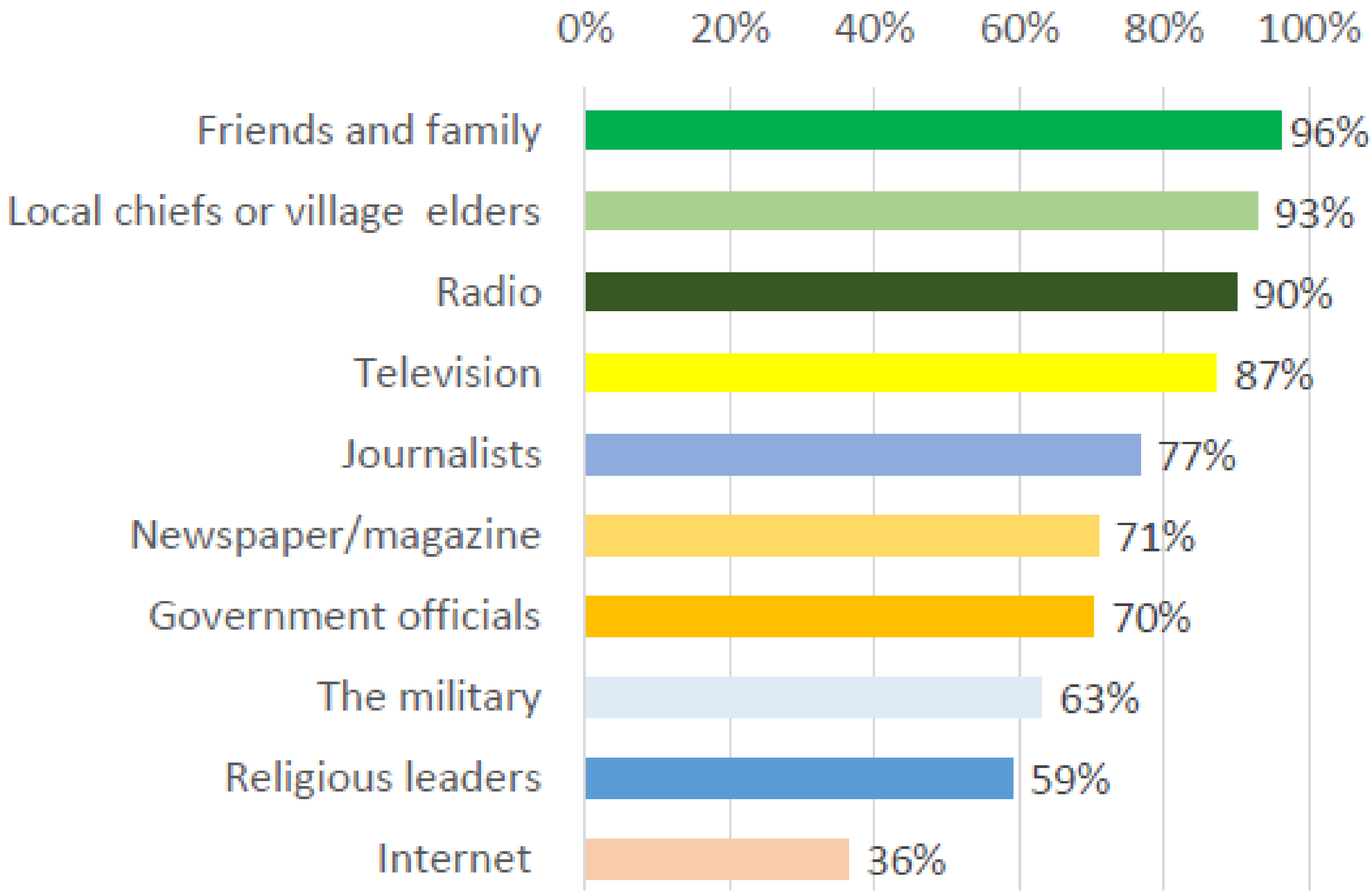


Sources of Information



Source: BBC Media Action, 2016

Most Trusted Information Sources



Source: BBC Media Action, 2016

Common Feedback Project (CFP)

- Community perceptions on reconstruction, livelihoods and food security, and protection on a regular basis
- Reports and data available on CFP report: http://www.cfp.org.np/
- Recent findings include:
  - Impact of language on access to information
  - Confusion around process to access reconstruction grant

FAQs

- http://www.hrrpnepal.org/faq

- Qs grouped under four themes:
  - Programme Setup,
  - GoN Housing Grant,
  - Housing Standards, and
  - FAQs from Households / Communities

- Qs can be submitted for inclusion to info@hrrpnepal.org or to the relevant HRRP district coordination team

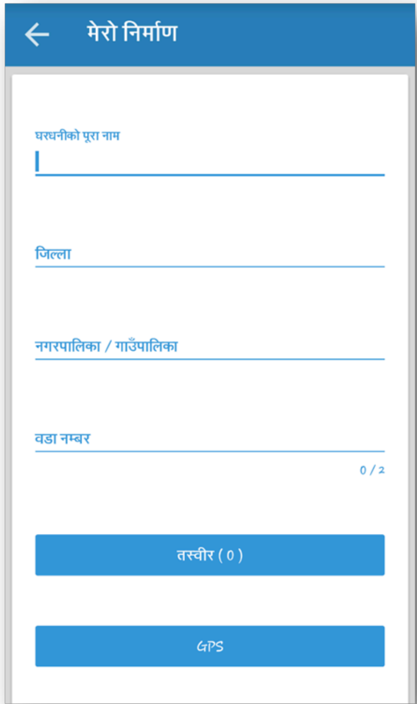
NRA Toll-free Number

- 16600172000 (१६६०–०१–७२०००) for NTC
- 9801572111 (९८०१५–७२१११) for Ncell

- 2 people are answering the phonelines
- Calls will be answered between 9-6, Sunday to Friday
- During off hours, questions will be recorded
- The staff answering the phonelines are based in the NRA and source information from NRA Communication team
- The number is supported by UNDP

Surakshit Ghar Mobile App

FAQs



My Project

Submit Qs

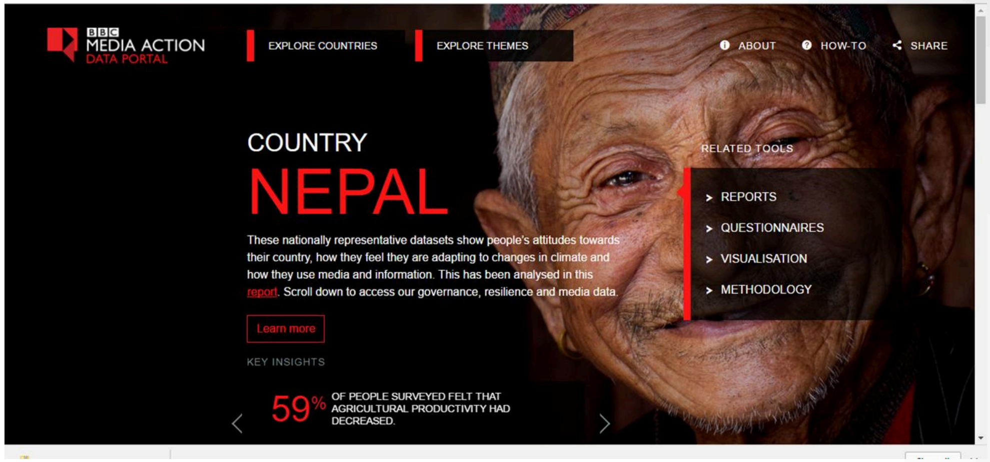
Contact

[App is available to download for free on the Google Play Store]()

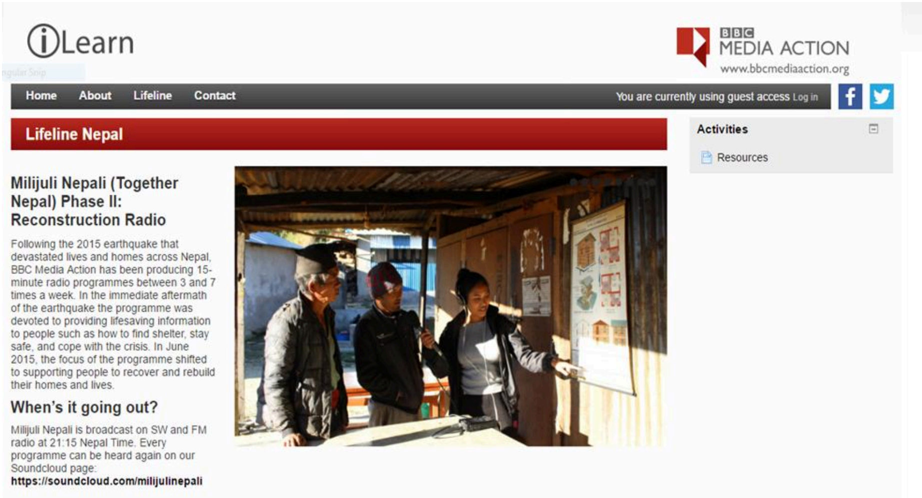
Housing Recovery & Reconstruction Platform (HRRP)

BBC Media Action

- [Data Portal]
- KathaMaala – radio drama



- [Millijuli Nepali] – radio programme
- [Dashboard]



Housing Recovery & Reconstruction Platform (HRRP)

USAID/NSET Baliyo ghar TV Program

Also Subscribe <u>Baliy Ghar</u> at Youtube Channel!

Get Involved!

- The HRRP website - http://hrrpnepal.org/
- The HRRP Flickr page (and share photos!) https://www.flickr.com/photos/hrrp/with/32476019890/
- The HRRP Facebook page - https://www.facebook.com/HRRPNepal/
- HRRP on Twitter - https://twitter.com/hrrp_nepal
- Share Qs for the FAQs to info@hrrpnepal.org

Thank You !